

HAOYU(ALVIS) YAN

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EDUCATION

Cornell University

August 2025 - May 2026

- Master of Engineering in Computer Science with a concentration in Machine Learning. GPA: 4.1/4.3

New York University

September 2021 - May 2025

- Bachelor of Arts in Computer Science and Economics. GPA: 3.9/4.0

PROFESSIONAL EXPERIENCE

Xenon Health

Jersey City, NJ

Software Engineer Intern | Hybrid, December 2025 - August 2026

- Built an **AI-OCR** system using a **React** front-end and **Quart** back-end to extract and process structured information from financial/billing PDF documents. Optimized the production **ML pipeline** with a parallel multi-stage filtering-extraction workflow, achieving over 95% accuracy on a 481-page test set. Deployed on **Microsoft Azure**.
- Built an **agentic medical coding module** (AIAMC), reduced manual review time by more than 50%.

Software Development Engineer | On-Site, August 2026 - present

- Built an end-to-end healthcare provider operations platform with a **React** front end, **Go** backend, and **RBAC**, supporting service scheduling, appointment confirmation, status tracking, invoice submission, and payment/charge tracking. Used by finance, billing, and recruiting teams and more than **200 anesthesia providers**..
- Developed a multi-channel agentic assistant system with **LangGraph**, integrated across web and email for provider support, service inquiries, status checks, and workflow actions via integrated APIs; deployed on **AWS**.

Utilyze, Inc.

United States

Machine Learning Engineer Intern | Remote, December 2025 - May 2026

- Led the predictive maps team to build an interactive **MapLibre** visualization platform and **data ingestion pipeline**, pulling and processing outage/infrastructure/weather data to enable real-time monitoring of 2,000+ critical infrastructure locations across service regions, deployed the MVP on **Vercel**.
- Developed a geospatial **ML pipeline** using NISAR L-band SAR imagery to investigate underground water leakage detection from remote-sensing signals. Deployed on **AWS**.

Bimini Vision/EyeSpace LLC

United States

Machine Learning Engineer Intern | Remote, April 2024 - February 2025

- Designed, developed and evaluated deep learning models with **PyTorch** and **Transfer Learning** to predict vault sizes (the space or distance between a contact lens or intra-ocular lens and the cornea) with OCT images (a non-invasive imaging scan of the eye) from ~200 patients. Achieved around 90% prediction accuracy within 400 microns for ResNet-based models.
- Developed and maintained **full-stack applications** with **Streamlit** for model hosting and demo.

PROJECT EXPERIENCE

Asterloom.ai

Project Developer | 2025 - 2026

- Designed and developed asterloom.ai, a full-stack AI writing platform for novel and script generation with versioned community content sharing, supporting both web and Android clients with responsive UI/UX.
- Built backend services and APIs using **FastAPI**, implementing AI-assisted writing workflows, versioned content management, and community features with **PostgreSQL/MongoDB, Redis, SSE streaming, and ARQ asynchronous jobs** following a hexagonal architecture. Deployed a cloud-native **AWS architecture**.
- Developed an Android client in **Kotlin**, implementing core application workflows for cross-platform user access.

Bond Fund Portfolio Management with Machine Learning

New York, NY

Research Assistant | May 2024 - May 2025

- Participated in a research project which apply Machine Learning in constructing mutual fund portfolios and making data-driven investment decisions.
- Constructed **deep learning models** in **PyTorch** to predict abnormal returns of bond funds using real-world **time series** financial data (1995 ~ 2023), built the **data pre-processing pipeline** using **Numpy** and **Pandas**.
- Evaluated model performance using financial metrics (sensitivity analysis, Sharpe ratio), visualized results using **Matplotlib**, and performed **data analysis** to identify investment opportunities and conduct risk assessments.

SKILLS

Languages: Python, C++, Go, JavaScript, SQL, R

ML/AI: PyTorch, TensorFlow, NumPy, Pandas, LangChain/LangGraph

Systems/Cloud: CUDA, Docker, Kubernetes, AWS, Azure, PostgreSQL

Web: React, Next.js, FastAPI, Quart, Streamlit

Relevant Courses: Advanced Machine Learning Systems, Cloud Computing&ML Hosting, Deep Learning, Algorithms